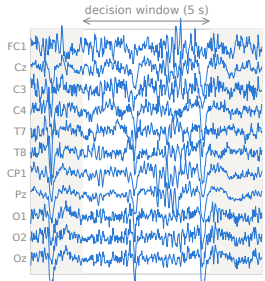
CoupleFlow

a candidate talker reaches the decision only through its time-locked coupling with the brain response

Neural path



EEG $\mathbf{x}$ (C channels × T samples)

Listener correction

$\mathbf{x} + U_\ell V_\ell^\top \mathbf{x}$

$U_\ell, V_\ell \in \mathbb{R}^{C \times 2}$

Virtual sensors

1×1 conv
$C \to 8$

Dilated encoder

sound at sample n

reads EEG $n \to n{+}0.98$ s

5 layers, dilation 1–16

$\mathbf{e} \in \mathbb{R}^{D \times T}$

neural embedding

$\mathbf{e}$

Flow head

conditional flow matching on the envelope

$R_c(a) = \|v_\theta(a_t, t, \mathbf{e}) - (a - \varepsilon)\|^2$

$R_u(a)$: the same with $c = \varnothing$

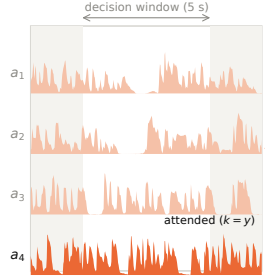
decision: $\mathrm{PMI}(a_k; \mathbf{x}) \approx R_u(a_k) - R_c(a_k)$

Time-centred coupling score

$$s_k = \frac{1}{\tau} \sum_{d=1}^{D} w_d \, \mathrm{corr}_t(e_d, \hat{a}_{k,d})$$

a time-constant $\mathbf{e}$ gives every $s_k = 0$

Acoustic path



candidate envelopes $a_1, \ldots, a_K$ ($K = 4$ here)

Candidate features

envelope, onsets,
8 + 28 sub-bands,
glimpsed / masked

Quantile match

equal value
distribution
across k

Shared audio encoder

causal dilated convs,
one set of weights
for every candidate

$\hat{\mathbf{a}}_k \in \mathbb{R}^{D \times T}$

candidate embeddings

reversed gradient

Training objective

$\mathcal{L} = \Sigma_j \lambda_j \mathcal{L}_j$

$\mathcal{L}_{\mathrm{CE}}$ s_k y — *decoding*

$-\log \mathrm{softmax}_k(s_k)[y]$

the attended candidate must win

$\mathcal{L}_{\mathrm{NCE}}$ $\mathbf{e}$ $\hat{\mathbf{a}}_k$ — *decoding*

$\frac{1}{2}[\mathrm{CE}(S, I) + \mathrm{CE}(S^\top, I)]$

$S_{ij} = s(\mathbf{e}_i, \hat{\mathbf{a}}_{j,y_j})$ over B windows of one listener

$\mathcal{L}_{\mathrm{FM}}$ $\mathbf{e}$ a_y — *density*

$\|v_\theta(a_t, t, c) - (a_y - \varepsilon)\|^2$

$a_t = (1-t)\varepsilon + t a_y$; $c = \mathbf{e}$, dropped w.p. 0.2

$\mathcal{L}_{\mathrm{null}}$ s_k s^{other} — *anti-shortcut*

$\mathrm{sp}(s_y^{\mathrm{other}} - s_y + \mu) + \mathrm{sp}(\mu - s_y)$

beat another window's EEG and a constant

$\mathcal{L}_{\mathrm{adv}}$ $\hat{\mathbf{a}}_k$ y — *anti-shortcut*

$\mathrm{CE}(g(\hat{\mathbf{a}}_{1:K}),\ y)$

audio-only classifier, gradient reversed

$\mathcal{L}_{\mathrm{VIC}}$ $\mathbf{e}$ — *anti-shortcut*

$\frac{1}{D}\Sigma_d [\gamma - \mathrm{sd}_t(e_d)]_+ + \frac{0.04}{D}\Sigma_{d \neq d'} \mathrm{Cov}_{dd'}^2$

keeps $\mathbf{e}$ varying and decorrelated

$\mathcal{L}_{\mathrm{side}}$ $\mathbf{x}$ side_y — *auxiliary*

$\mathrm{CE}(h(\mathbf{x}),\ \mathrm{side}_y)$

attended side from EEG alone

$\mathcal{L}_{\mathrm{env}}$ $\mathbf{e}$ a_y — *auxiliary*

$1 - r(\hat{\mathbf{w}}^\top \mathbf{e},\ a_y)$

linear envelope read-out

Symbols

$\mathbf{x}$: EEG window incl. context, C channels × T samples (64 Hz) ℓ: listener U_ℓ, V_ℓ: rank-2 channel correction of listener ℓ $\mathbf{e}$ / e_d: neural embedding / its dimension d ($D = 16$)

a_k: envelope of candidate talker $k = 1..K$ $\hat{\mathbf{a}}_k$ / $\hat{a}_{k,d}$: its embedding / dimension d y: index of the attended talker corr_t: Pearson correlation over time w_d: learned weights

τ: learned temperature B: windows in a batch (all from one listener) I: identity (matching pairs) s_y^{other}: s_y computed with another window's EEG sp: softplus $\mu = 0.5$: margin

g: audio-only classifier $\gamma = 0.5$: minimum temporal s.d. sd_t: s.d. over time $\mathrm{Cov}_{dd'}$: covariance of $\mathbf{e}$ over time v_θ: flow velocity network $t \sim \mathcal{U}(0, 1)$: flow time

$\varepsilon \sim \mathcal{N}(0, I)$: noise c: condition ($\varnothing$: dropped) R_c, R_u: conditional / unconditional residual h: side head side_y: side of talker y $\hat{\mathbf{w}}$: linear read-out r: Pearson correlation

CE: cross-entropy $\mathrm{softmax}_k$: normalised over the K candidates $[\cdot]_+ = \max(\cdot, 0)$ PMI: pointwise mutual information $\log p(a|\mathbf{x}) - \log p(a)$ n: sample index

λ_j: NCE 1, null 0.5, adv 0.3, FM 0.3, side 0.3, VIC 0.1, env 0.2 (CE 1) Signals: MAESTRO listener S03, 7 s (5 s window + 1.56 s context each side), attended talker a_4